

Module 1: Introduction to Problem Solving & Complexity

(Basics of algorithms, time & space complexity, basic I/O, logic building)

1. Two Sum

 LeetCode #1

Concepts: Brute force vs optimized approach, time complexity

Why: Classic problem to understand $O(n^2)$ vs $O(n)$

2. Running Sum of 1D Array

 LeetCode #1480

Concepts: Basic algorithm, prefix idea

Why: Simple logic + iteration practice

3. Number of Steps to Reduce a Number to Zero

 LeetCode #1342

Concepts: Algorithm design, loops, condition checking

Why: Helps with step-by-step problem solving

4. Maximum Number of Words Found in Sentences

 LeetCode #2114

Concepts: Basic input handling, counting logic

Why: Good for understanding problem constraints

5. Find the Highest Altitude

 LeetCode #1732

Concepts: Prefix sum, space optimization

Why: Introduces optimization thinking early

6. Palindrome Number

 LeetCode #9

Concepts: Algorithm logic, condition checking

Why: Trains step-by-step logical thinking

7. Fizz Buzz

 LeetCode #412

Concepts: Basic algorithm, loops, conditionals

Why: Very good for beginners and logic building

8. Power of Two

🔗 LeetCode #231

Concepts: Optimization, bit manipulation, time complexity

Why: Shows efficient vs brute force approach

9. Count Odd Numbers in an Interval Range

🔗 LeetCode #1523

Concepts: Mathematical optimization, $O(1)$ solution

Why: Excellent example of time optimization

10. Find Numbers with Even Number of Digits

🔗 LeetCode #1295

Concepts: Looping, basic math, complexity analysis

Why: Simple but reinforces counting logic